

Binxin Gao

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Education

University of Maryland, Robert H. Smith School of Business Aug 2024-present
Ph.D. in Management, College Park
Independent Project: *"OpenAI or CloseAI? That is A Question: A Model & Some Empirical Evidences"*
Mentors: Prof. Serguey Braguinsky and Prof. Evan Starr

Columbia University, Columbia Business School May 2016
M.Sc., Financial Economics, Core GPA: 3.7/4.0 New York
Thesis: *"Does Shanghai-Hong Kong Stock Connect really connect A-share and H-share market?"*
Mentor: Prof. Michael Johannes

Peking University, Guanghua School of Management July 2014
B.Sc., Finance & Statistics, Core GPA: 3.75/4.0 Beijing
Honors Thesis: *"Non-linear Momentum Trading Strategies in Chinese Future Market"*
Mentor: Prof. Chenxu Li

Fundings

- OpenAI Research Access Program

Ongoing Work

Gao, B. *"OpenAI or CloseAI? That is A Question: A Model & Some Empirical Evidences"*. (Partially supported by OpenAI Research Access Program)
Gao, B. *"IP to IPO: Patent Purchases and the Market for Window Dressing"*.

Publications

Gao, B., & Han, J. (2025). *Max It or Miss It: Benchmarking LLM On Solving Extremal Problems*, arXiv preprint arXiv:2510.12997.
Gao, B., Li, C., Qiao, K. (2015). *Non-linear Momentum Strategies: Theories and Applications in the Chinese Future Market*, Quarterly Journal of Finance, 9(1), pp. 23–58.
○ 11 citations, 325 downloads.
○ Recognized by top Chinese universities for tenure requirements.

Service

- **Reviewer:** NeurIPS 2025.

Professional Experience

CCB International (China) Limited, Hangzhou
Vice President, Fund Management Department

June 2022 – April 2024

- Led deal origination and execution in technology and consumer sectors, managing an investment portfolio across AI, SaaS, optoelectronics, etc.
- Closed 3 transactions worth ¥180M, mentoring associates and overseeing valuation modeling, due diligence, and SPA negotiation.
- Cultivated relationships with LPs, local government, and PE/VC networks to source over 100 partnership opportunities.

Tencent, Shanghai/Shenzhen
Product Expert, Corporate Development Group (AMS)

Nov 2020 – Apr 2022

- Sourced and evaluated 20+ strategic partnerships in consumer and tech sectors, including XR and IP products.
- Built AI tools leveraging CV and NLP algorithms, improving advertiser ROI for key clients (total budget: ¥50M).
- Conducted A/B tests, authored product requirement documents, and advised clients on strategies.

NetEase Games, Hangzhou
Senior Game Designer, Interactive Entertainment Business Group

July 2016 – Sep 2019

- Designed core features for the MMORPG “Chuliuxiang,” achieving 10M downloads and 3M DAU.
- Led cross-functional teams to create innovative gameplay elements, such as terrain interaction and auction systems.
- Founded a Hanfu brand and collaborated with Dunhuang Research Institute on 16 original designs, garnering 10k+ followers.

Research Experience

Thesis Seminar at Columbia Business School
supervised by Prof. Michael Johannes (chair of MSFE program)

Feb 2016 – May 2016

- Assessed Shanghai-Hong Kong Connect’s impact on market efficiency using Difference-in-Differences and propensity score matching.
- Developed quantitative research skills across research design, data collection, analysis, and presentation.

Behavioral Finance Project
supervised by Prof. Chenxu Li, Ya Tang, Lei Lu at Peking University

Sep 2013 – Sep 2014

- Proposed a model drawing on financial intuitions by substituting “binary bet” with a double-step function, which represents waiting until the momentum signal reaches a threshold before trading, and introducing some hidden factors like Shibor
- Investigated all 660 contracts in 2012 thoroughly using MATLAB, located and elucidated the non-linear momentum effects in the Chinese future market as the first foray to fill in the blanks in both theory and industry

- Constructed trading strategies that significantly outperformed the traditional linear approach with double excess return and significantly improved Sharpe, and checked the robustness with trading data in 2013
- Independently completed the paper "Applications of Non-linear Momentum Trading Strategies in Chinese Future Market"

Stochastic Process Textbook Project

Sep 2013 – Jun 2014

Head Research Assistant at Guanghua School of Management

- Co-authored the textbook "Applied Mathematical Methods in Finance Research" (ISBN: 978-7-301-31844-7) using LaTeX
- Organized group seminars and authored introductory documents on LaTeX for group members
- Replicated (Martin & Bana, 2012) in Matlab and SAS by collecting, cleaning, and analyzing high-frequency trading data

CBOE Market Volatility Index (VIX) Project

Sep 2013 – Dec 2013

Research Assistant for Prof. Chenxu Li at Peking University

- Co-authored a working paper on the pricing of VIX derivatives under a general class of stochastic volatility models by combining the Ito-Taylor expansion with a Markov chain Monte Carlo simulation
- Conducted literature review on the term structure of implied volatility and attended conferences (eg. Jim Gatheral's talk)

Honors & Awards

- **Province Award of Excellence (2015):** Stipend awarded to top-performing overseas students.
- **Photography (2013):** 1st Prize out of 1,178,307 in National Photography Competition.
- **National Business Case Analysis Contest (2013):** Best Individual and Team Awards.

Skills

Research Methods: DID, RDD, IV, SEM, Fixed/Random Effects, Case Studies, Survey Design.

Programming Languages: Python, R, VBA, MATLAB, SQL, C++.

Software: Stata, Bloomberg, SAS, SPSS, Capital IQ.

Languages: English (TOEFL 112, GMAT 770), Mandarin (Native).

Interests

Hanfu vlogger with 60k+ followers, Tea ceremony, snowboarding, photography, and calligraphy.